

DAYANANDA SAGAR COLLEGE OF ENGINEERING

Shavige Malleshwara Hills, Kumaraswamy Layout, Bangalore-560078

Department of Artificial Intelligence and Machine Learning



INTERNSHIP REPORT

ON

“Flights API”

Submitted in partial fulfillment for the award of degree(20AI8ICINT)

BACHELOR OF ENGINEERING IN

Department of Artificial Intelligence and Machine Learning

Submitted by:

C.Tejashwini - 1DS20AI019

Conducted at



SmartBridge

Department of Artificial Intelligence and Machine Learning

Dayananda Sagar College of Engineering Bangalore-78

2023-2024

DAYANANDA SAGAR COLLEGE OF ENGINEERING

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CERTIFICATE

This is to certify that the Internship titled “**Flights API**” carried out by **Ms. C.TEJASHWINI [1DS20AI019]**, a bonafide student of Dayananda Sagar College of Engineering, in partial fulfillment for the award of **Bachelor of Engineering**, in **Artificial Intelligence and Machine Learning** during the year 2023-2024. It is certified that all corrections/suggestions indicated have been incorporated in the report.

The internship report has been approved as it satisfies the academic requirements in respect course Internship (20AI8ICINT)

Signature of Reporting Head
Swathi Katikaneni
Team Lead
SmartBridge

Signature of Internship Coordinator
Prof. Kavya D N
Dept. of AI & ML
DSCE

Signature of HoD
Dr. Vindhya P Malagi
Dept. of AI & ML
DSCE

External Viva:

Name of the Examiner

1) _____

2) _____

Signature with Date

DAYANANDA SAGAR COLLEGE OF ENGINEERING

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Department of Artificial Intelligence and Machine Learning



DECLARATION

I, **C.TEJASHWINI**, final year student of Information Science and Engineering, Dayananda Sagar College Of Engineering, Bangalore - 560 078, declare that the Internship has been successfully completed, in **SMARTBRIDGE**. This report is submitted in partial fulfillment of the requirements for the award of Bachelor Degree in Artificial Intelligence and Machine Learning, during the academic year 2023-2024.

Date:

USN: 1DS20AI019

Place: Bangalore

NAME: C.Tejashwini

OFFER LETTER



Welcome to SmartInternz! Delighted to have you on board 😊

1 message

SmartInternz <info@smartinternz.com>
Reply to: Smart bridge <info@thesmartbridge.com>
To: tejeshwinichalla.28@gmail.com

Fri, 2 Jun, 2023 at 8:20 pm



Dear C.Tejashwini ,

Greetings from SmartInternz!

You have successfully enrolled for the Salesforce Supported Virtual Internship Program 2023 - Mulesoft Track!

To know about the event format and timelines - [Click here](#)

Please find the handbook to know the step by step procedure to complete course modules - [Click here](#)

Login to your Student account to start your Virtual Internship Program.

Username : tejeshwinichalla.28@gmail.com

Password : 6U3cYf0q

Login

Thanks,

[SmartInternz](#)

COMPLETION CERTIFICATE



In Partnership With



CERTIFICATE OF COMPLETION

October 05, 2023

C. Tejashwini

Mulesoft Certified Developer - Level 1

During the 8 Weeks period of Virtual Internship (**June-August 2023**), C. Tejashwini has completed the following modules from Mulesoft Anypoint Platform Development Fundamentals

Getting Started With Any Point Platform

Application Networks & API-Led Connectivity
Anypoint Platform Core Components
Efficient API Design
Robust API Building & Deployment

Building Mule Applications With Any Point Studio

Mule Event Manipulation & Scalability
Web Service Integration & Data Management
Error Handling & DataWeave Transformations
Flow Automation & Record Processing

Certificate ID: SISFVIPAD2023 -64460 | Verify this certificate @ https://smartinternz.com/internships/salesforce_certificates/ab34d8d7f2771066c8724c8c4d440f60

Shri Buddha Chandraseker

Chief Coordinating Officer(CCO),
NEAT Cell-AICTE

Mr Amarender Katkam

Founder & CEO, TheSmartBridge &
SmartInternz

ACKNOWLEDGEMENT

This Internship is a result of accumulated guidance, direction and support of several important persons. We take this opportunity to express our gratitude to all who have helped us to complete the Internship.

We express our sincere thanks to our Principal **Dr. B G Prasad**, for providing us adequate facilities to undertake this Internship.

We would like to thank **Dr. Vindhya P Malagi**, Head of Department – AI & ML, for providing us an opportunity to carry out Internship and for her valuable guidance and support.

We would like to thank **Swathi Katikaneni**, SmartBridge, for guiding us during the period of internship.

We express our deep and profound gratitude to our Internship Co-ordinator, **Prof. Kavya D N**, Assistant Professor – AI & ML, for her keen interest and encouragement at every step in completing the Internship.

We would like to thank all the faculty members of AI & ML department for the support extended during the course of Internship.

We would like to thank the non-teaching members of the Department of AI & ML, for helping us during the Internship.

Last but not the least, we would like to thank our parents and friends without whose constant help, the completion of Internship would have not been possible.

C.TEJASHWINI [1DS20AI019]

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CHAPTER 1

INTRODUCTION

During my 8-week virtual internship at SmartBridge, in collaboration with MuleSoft, I gained hands-on experience in utilizing the Anypoint Platform™ for API development and integration tasks. The internship primarily focused on two objectives: leveraging the capabilities of the Anypoint Platform™ for comprehensive API lifecycle management, including discovery, design, building, deployment, management, and governance, and refining skills in building Mule applications with the Anypoint Platform™.

My time at Smart Bridge as a virtual intern in the role of a MuleSoft developer was an immersive dive into the realm of digital integration and innovation. Over the course of eight weeks, I had the privilege of contributing to real-world projects, honing my skills, and collaborating with a team of industry experts. Smart Bridge provided a dynamic environment where learning was paramount, fostering a culture of curiosity and creativity that propelled my professional growth. In this report, I offer insights into my journey, highlighting key experiences, challenges, and achievements during my tenure at this forefront company in the field of digital transformation.

CHAPTER 2

DESCRIPTION OF ORGANIZATION

SmartBridge is at the forefront of revolutionizing talent acquisition and development within the rapidly evolving technology landscape. Dedicated to cultivating a next-generation talent pool proficient in the high-demand technologies shaping industry innovation, SmartBridge employs a distinctive approach known as "Project-based learning." This approach is forged through collaboration with leading global technology firms, bridging the gap between academic knowledge and industry application.

Central to SmartBridge's mission is the establishment of a symbiotic relationship between industry and academia, facilitating the seamless transition of students into the workforce. The platform offers students unparalleled opportunities to engage in virtual internships with top-tier companies, providing practical, hands-on experience in cutting-edge technologies. This experiential learning model ensures that graduates emerge equipped with the skills and expertise demanded by today's job market.

SmartBridge champions the concept of a demand-driven talent pool as a proactive strategy for aligning workforce capabilities with industry needs. By strategically identifying and nurturing skilled professionals, SmartBridge empowers organizations to swiftly adapt to evolving market demands, enhancing operational agility and gaining a competitive advantage. The commitment to excellence ensures that businesses have access to the right talent at the right time, propelling them towards sustained success and fostering innovation in the technology sector.

CHAPTER 3

SCOPE OF WORK

During my internship at SmartBridge, I was entrusted with a diverse array of responsibilities, primarily centered around harnessing the capabilities of the Anypoint Platform™ for API development and integration tasks. The scope of my work revolved around two key objectives.

Firstly, I gained hands-on experience in leveraging the Anypoint Platform™ for comprehensive API lifecycle management. This involved every aspect from initial discovery and design through to building, deployment, management, and governance. Through practical application and guided learning, I developed proficiency in navigating the various stages of the API lifecycle, ensuring that APIs were efficiently managed and optimized for performance and scalability.

Secondly, the internship afforded me the opportunity to refine my skills in building Mule applications with the Anypoint Platform™. This aspect of my work focused on practical implementation, where I translated theoretical knowledge into tangible solutions. By building Mule applications, I honed my ability to architect and develop robust integration solutions, leveraging the full potential of the Anypoint Platform™ to seamlessly connect disparate systems and data sources.

Throughout these tasks, I collaborated closely with mentors and peers, seeking guidance when needed and sharing insights gained from hands-on experience. Additionally, I actively participated in workshops and training sessions provided by SmartBridge, further enhancing my understanding of the Anypoint Platform™ and its capabilities.

CHAPTER 4

LEARNING OBJECTIVES AS GIVEN BY COMPANY

- **Building an Application Network:** Gain practical experience in constructing an application network using API-led connectivity principles and the Anypoint Platform. Learn how to design interconnected systems that facilitate seamless communication and data exchange.
- **Utilizing Anypoint Platform for API Lifecycle Management:** Develop proficiency in utilizing the Anypoint Platform for API lifecycle management, including discovery, design, building, deployment, management, and governance. Ensure the efficiency, reliability, and scalability of APIs within the application network.
- **Connecting to Data Sources and Services:** Acquire skills in establishing connections to various data sources and services, such as databases, files, web services, and SaaS applications. Learn how to integrate disparate systems and leverage data from multiple sources to enhance the functionality of integration solutions.
- **Implementing Application Logic and Data Transformation:** Gain hands-on experience in adding application logic, error handling, and data transformation capabilities using DataWeave. Learn how to manipulate and transform data formats like XML, CSV, and JSON to meet integration solution requirements.
- **Structuring Applications for Development and Deployment:** Learn best practices for structuring applications effectively to facilitate streamlined development and deployment processes. Understand how to organize projects, manage dependencies, and ensure scalability and maintainability.
- **Handling Batch Data Processing:** Develop skills in efficiently handling batch data processing tasks using the Anypoint Platform. Gain an understanding of batch processing concepts, techniques, and best practices to effectively manage large volumes of data within integration solutions.

CHAPTER 5

CHALLENGES AND SOLUTIONS

As an intern at SmartBridge, I was assigned to work on the American Flights API project, which facilitates operations on the "American" table within the training database. The API, developed in Anypoint Studio and deployed on the Anypoint Platform, plays a pivotal role in retrieving, adding, deleting, and updating flights. Leveraging the platform's API Designer, the API was meticulously designed to seamlessly integrate with CloudHub for efficient deployment.

Challenges and solutions :

1. Data Transformation and Formatting:

- Challenge: The primary challenge was to transform data into the appropriate format before delivering it to designated systems.
- Solution: Comprehensive data transformation pipelines were implemented using DataWeave, ensuring seamless compatibility with various data sources and systems.

2. Integration with Web Services:

- Challenge: Integrating with external web services required establishing secure and reliable connections.
- Solution: Mule's robust capabilities were utilized to integrate with web services securely, implementing techniques such as API rate limiting and caching to optimize performance.

3. Error Handling Mechanisms:

- Challenge: Implementing robust error handling mechanisms was essential to ensure the reliability of the API.
- Solution: A robust error handling framework was developed within Mule applications, including retry mechanisms and fault tolerance strategies.

4. Efficient Record Processing:

- Challenge: Efficiently handling and processing records within Mule applications, particularly in the Salesforce database, was challenging.
- Solution: Efficient record processing techniques were employed, leveraging batch processing capabilities and optimizing data retrieval and manipulation workflows.

CHAPTER 6

ACHIEVMENTS AND CONTRIBUTION

Designing API for Flights

The American Flights API facilitates operations on the "American" table within the training database. It supports functions like retrieving all flights, accessing specific flights by ID, adding, deleting, and updating flights. Developed in Anypoint Studio and deployed on the Anypoint Platform, the API was designed using the platform's API Designer. It's seamlessly integrated with CloudHub for efficient deployment

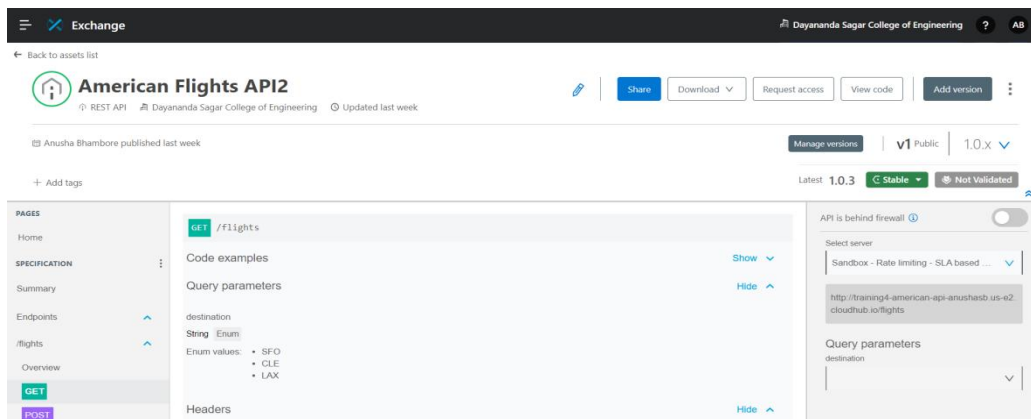


Fig 6.1: Anypoint Platform API Portal in Exchange

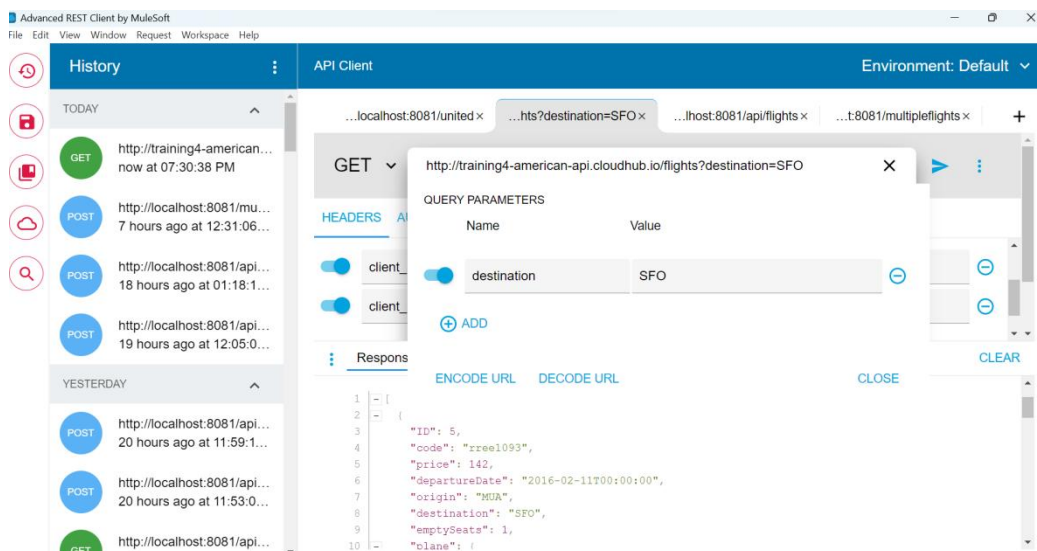


Fig 6.1: Advanced Rest Client

Building Mule Application

Mule applications were carefully designed to connect different systems, making it easy to move data between them. The main task assigned was to read, process, and change data into the right format before delivering it to designated systems. Some tasks carried out throughout the internship are:

- Accessing and Modifying Mule Events: Working with Mule events to handle data flow.

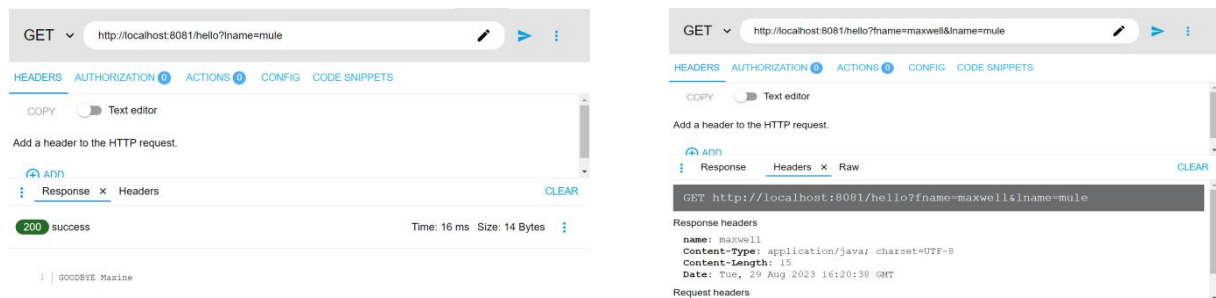


Fig 6.3: Accessing and Modifying Mule Events

- Consuming Web Services and Controlling Event Flow: Integrating with web services and managing event flow in Mule applications.

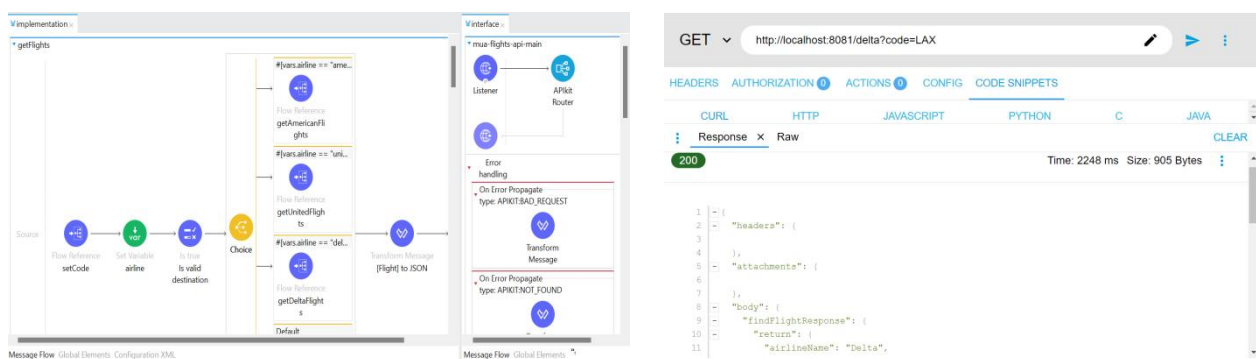


Fig 6.4: Consuming Web Services and Controlling Event Flow

- Handling Errors: Implementing error handling mechanisms in Mule applications.

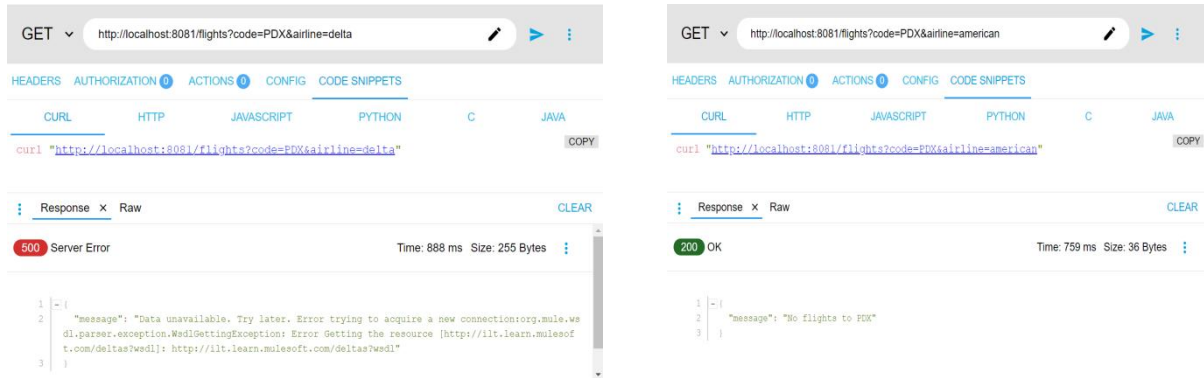


Fig 6.5: Handling Errors

- Processing Records: Handling and processing records efficiently within Mule applications in the salesforce database.

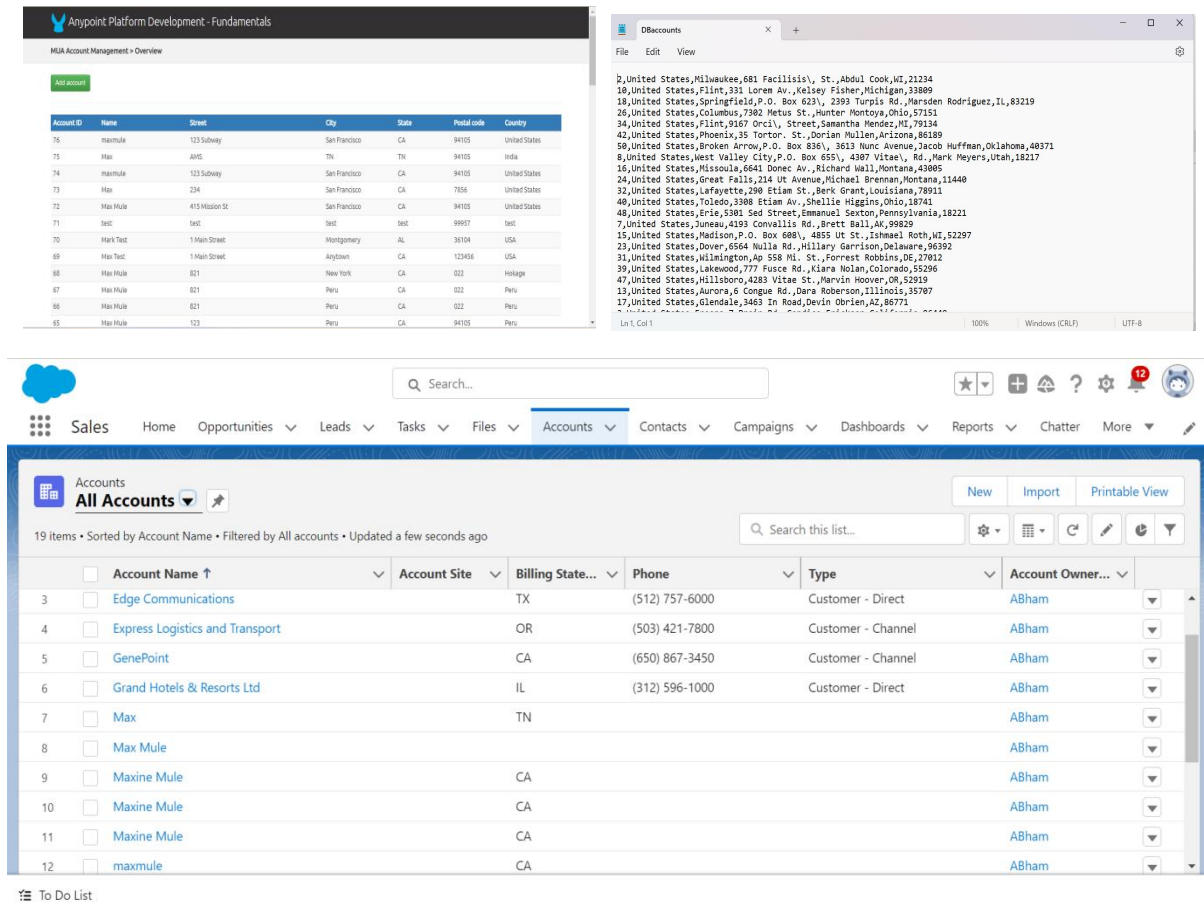


Fig 6.7: Processing Records

CHAPTER 7

REFLECTION AND ANALYSIS

My internship at SmartBridge was an enriching journey that provided me with a multifaceted learning experience. Working extensively with MuleSoft integration, I gained a deep understanding of APIs, integration patterns, and the intricacies of the Anypoint Platform. From designing to deploying APIs and managing integrations, I was immersed in the technical intricacies that underpin modern digital connectivity. This hands-on experience not only solidified my technical proficiency but also equipped me with the skills necessary to tackle real-world integration challenges effectively.

Collaborating within SmartBridge teams and engaging directly with clients was a pivotal aspect of my internship experience. Through these interactions, I honed my project management and communication skills, learning how to navigate complex project scopes, gather comprehensive requirements, and present tailored solutions to meet clients' specific needs. Engaging in client-facing roles allowed me to understand the importance of effective communication and building strong relationships in delivering successful projects. Moreover, troubleshooting issues in real-time provided me with invaluable experience in agile problem-solving, a skillset essential for any professional endeavor.

Being part of the Mulesoft ecosystem provided me with unique insights into the broader industry landscape. Understanding how MuleSoft integrates seamlessly with Salesforce CRM and Marketing Cloud expanded my perspective on the potential of integration technologies in driving business innovation and efficiency. Additionally, networking opportunities within the Salesforce and MuleSoft communities allowed me to forge connections with industry professionals and gain exposure to emerging trends and best practices. Overall, my internship at SmartBridge not only deepened my technical expertise but also equipped me with invaluable soft skills and industry insights that will undoubtedly shape my future career trajectory.

CHAPTER 8

RECOMMENDATIONS



Dear C Tejashwini,

Based on the insights gained during your internship at SmartBridge, we highly recommend you as a candidate who brings a unique blend of technical proficiency, project management acumen, and client engagement skills to any team or project.

Throughout your internship, you demonstrated a remarkable ability to grasp complex technical concepts and apply them in real-world scenarios. Your hands-on experience with MuleSoft integration, including designing, developing, and managing APIs, showcases your aptitude for navigating the intricacies of modern digital connectivity.

Overall, I confidently recommend you as a candidate who not only brings technical expertise but also possesses the interpersonal skills and industry insights necessary to drive success in any endeavor. Your dedication, adaptability, and passion for innovation make you an asset to any organization.

Regards,

Team SmartBridge!

CHAPTER 9

CONCLUSION

My internship experience at SmartBridge has not only provided me with technical skills but has also instilled in me a strong sense of adaptability and resilience in dynamic work environments. Throughout various challenges and opportunities encountered during the internship, I have consistently demonstrated the ability to learn quickly and thrive in fast-paced settings. This adaptability, coupled with my continuous drive for improvement, positions me as a candidate capable of tackling new challenges with enthusiasm and determination.

Looking ahead, I am committed to furthering my professional growth and making meaningful contributions to the organizations and communities I engage with. Whether it's through continued learning, mentorship, or active participation in industry events, I am dedicated to staying at the forefront of technological advancements and best practices. My internship experience has not only been a stepping stone in my career but has also laid a solid foundation for future success.

In summary, my internship journey at SmartBridge has been a transformative experience that has shaped me into a well-rounded professional ready to take on the challenges of tomorrow. With a strong foundation in technical expertise, project management skills, and a client-centric approach, I am poised to make significant contributions in any role I undertake. I am grateful for the opportunities afforded to me during this internship and excited about the possibilities that lie ahead as I continue to grow and evolve in my career.

CHAPTER 10

REFERENCES

- [1] https://github.com/28042002/Mulesoft_virtual_internship
- [2] https://github.com/28042002/Mulesoft_virtual_internship/tree/main/AmericanFlightsAPI
- [3] https://github.com/28042002/Mulesoft_virtual_internship/tree/main/Mule%20Applications
- [4] <https://www.thesmartbridge.com/>
- [5] <https://www.salesforce.com/in/>
- [6] <https://www.mulesoft.com/>
- [7] <https://smartbridge.teachable.com/courses/enrolled/2110867>
- [8] <https://smartinternz.com/>
- [9] <https://anypoint.mulesoft.com/exchange/portals/dayananda-sagar-college-33/>
- [10] <http://training4-american-api-anushasb.us-e2.cloudhub.io/flights>
- [11] <https://anypoint.mulesoft.com/>

